

Herringbone Switchback

A3 booklet / Planet-300894

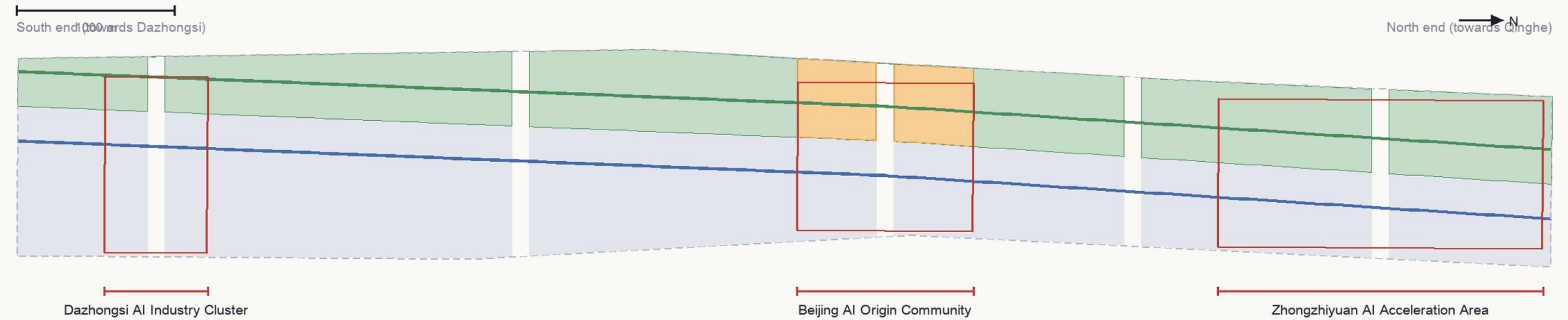
In 1899, when the Jing-Zhang Railway was surveyed, the grade through the Guangou pass convinced everyone the line could not be built. What finally solved it was not a larger locomotive but a reversal: at Qinglongqiao two legs of track meet at a single point, and by backing up, a train climbs. The herringbone was never ornament. It was a structural concession forced by terrain, and the concession became the most remembered part of the railway.

More than a hundred and twenty years later the same corridor must carry an artificial-intelligence innovation belt. Its shape is just as demanding: roughly 9.7 km north to south and only about 1.37 km east to west, a ratio approaching seven to one. Such a proportion admits neither radial centrality nor a grid; any scheme that places a single core at the middle immediately finds both ends torn away. This proposal therefore reuses the reversal - not as a figure drawn on the ground, but as an actual two-line corridor: a continuous heritage green spine to the west and a scenario-empowerment belt to the east, each standing on its own, meeting and exchanging dominance at the Beijing AI Origin Community, then continuing to both ends. Retreat to advance; two lines in parallel; one point of engagement.

Recomputed design area 11,412,825 m² 0.1125% from the announced 11.4 M sq m	Key areas total 369.3 ha cross-checked against 368.4 ha	Green ratio 34.0% a design outcome; no official control supplied	Public space share 10.0% apex plaza plus five public corridors
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Herringbone Switchback: Overall Spatial Structure

Centennial Jing-Zhang AI Innovation Belt · overall design area approx. 11.4 sq km · a 9.7 km by 1.37 km corridor



Concept origin: the Qinglongqiao herringbone

West: green spine

East: scenario belt

Apex

In 1909 the Jing-Zhang railway beat the Guangou grade with a herringbone switchback: two legs let a train reverse to climb. The plan translates that logic into two lines that stand on their own and swap roles at a single apex.

- Legend
- West line: heritage park green spine
 - Apex plaza (plaza land)
 - East line: Xiaoyuehe scenario belt
 - West slow-mobility line
 - East scenario service line
 - Three key areas (provisional)
 - Provisional rough boundary (dashed)

Core metrics (recomputed from geometry)	
Overall design area	11,412,825 m ²
Announced text value	11,400,000 m ²
Green ratio	34.0%
Public space ratio	10.0%
Road land ratio	5.6%
Slow-mobility line length	25.3 km
Land-use partition residual	22.4 m ²
Floor area ratio, height and density controls were not supplied with the site package; metrics.json declares them pending rather than inventing values.	

Data: geometry from geometry/*.geojson, areas recomputed in EPSG:4548; boundary from the maintainer-registered provisional rough boundary brief/site-package/geometry/provisional_boundaries.geojson. The map is rotated 90 degrees clockwise; see the north arrow. Dashed lines are a provisional constraint. They are not an official redline, an approval basis, or a precise-area basis; recompute everything once official polygons are released. An open co-creation concept suggestion. It does not replace statutory planning; floor area ratio, height, demolition decisions and road redlines are outside its conclusions.

One non-straight divider splits the 11.4 sq km corridor into a west heritage green spine and an east scenario belt; the two lines reach equal width at the AI Origin Community, forming the reversal apex. Five transverse stitch corridors connect the two sides.

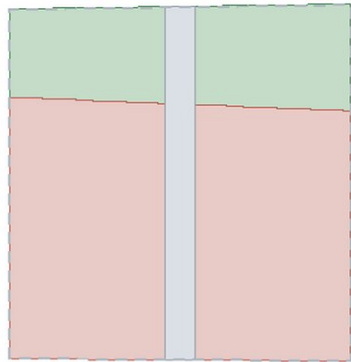
Land-use Structure: A Complete Partition

Six land-use classes tile the entire design area · partition residual 22.4 sq m (about 0.0002% of the total)



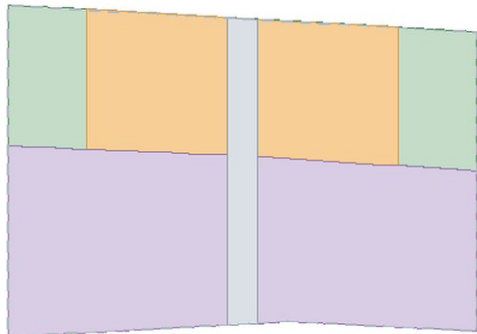
dazhongsi ai industry cluster

South: service-led, commercial frontage on the east



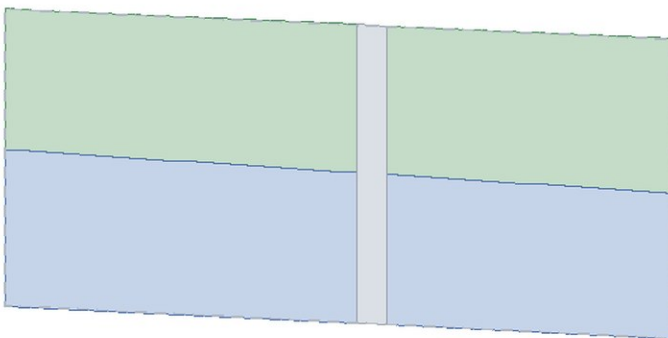
beijing ai origin community

Middle: the apex, plaza land bridges both lines



zhongzhiyuan ai acceleration area

North: research-led, research land parallels the spine



Land-use classes and recomputed areas (national classification codes)

Park green (1401) · 381.3 ha	Urban residential (0701) · 235.9 ha
Plaza (1403) · 49.9 ha	Commercial service (05) · 265.2 ha
Research (0802) · 145.1 ha	Road land (1207) · 63.8 ha

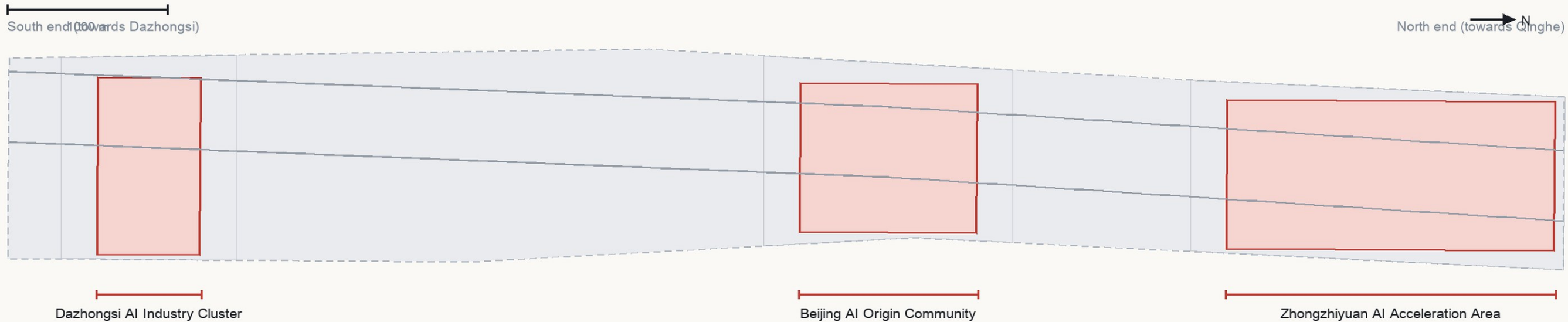
The partition is cut by one shared divider, so adjacent parcels use identical edge coordinates and leave no gaps or overlaps; total land-use area equals the site area. The classes are a concept suggestion; floor area ratio, height and demolition decisions are statutory judgements for professional teams to resolve.

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Six land-use classes tile the design area completely; adjacent parcels share identical edge vertices snapped to a 1 cm grid, leaving a residual of only 22.4 sq m against the site area as evidence of topological closure.

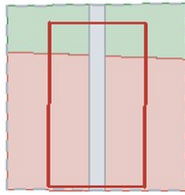
Key Areas: Three Areas and Two Wings

Three key areas total approx. 369.3 ha recomputed · announced text value approx. 368.4 ha



Dazhongsi AI Industry Cluster

AI-native business formats · south descending leg



Next to existing rail and a commercial hinterland. The suggestion is to lead with AI-native consumption, business services and low-speed autonomous delivery, converting northern innovation into market outcomes behind a continuous active frontage.

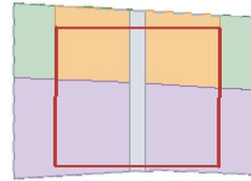
Recomputed 72.0 ha / announced approx. 72.0 ha

Provisional extent, not a precise-area basis

- Key area extent (provisional rough)
- Rest of the overall design area

Beijing AI Origin Community

World-class AI ecosystem · the apex



The apex where the herringbone reverses, and the spatial and narrative core. West spine and east scenario belt meet and swap dominance; the plaza carries public AI experience, developer community events and honour displays, ringed by housing and community services.

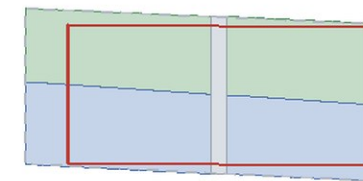
Recomputed 104.3 ha / announced approx. 104.3 ha

Provisional extent, not a precise-area basis

The two wings run the length of the belt rather than occupying separate parcels: the Zhongguancun service wing places knowledge and service nodes along the west spine, and the Xiaoyuehe scenario wing organises testable living and industrial scenarios along the east line. They meet at the apex plaza, so the three areas read as one walkable innovation chain rather than three islands.

Zhongzhiyuan AI Acceleration Area

Full-stack indigenous AI · governance voice · north climbing leg



Research-led, hosting full-stack indigenous R&D together with standards, evaluation and governance rule-making. Open testbeds, evaluation facilities and pilot-production space are suggested for further study, set against the spine so they share its continuous open space.

Recomputed 192.9 ha / announced approx. 192.1 ha

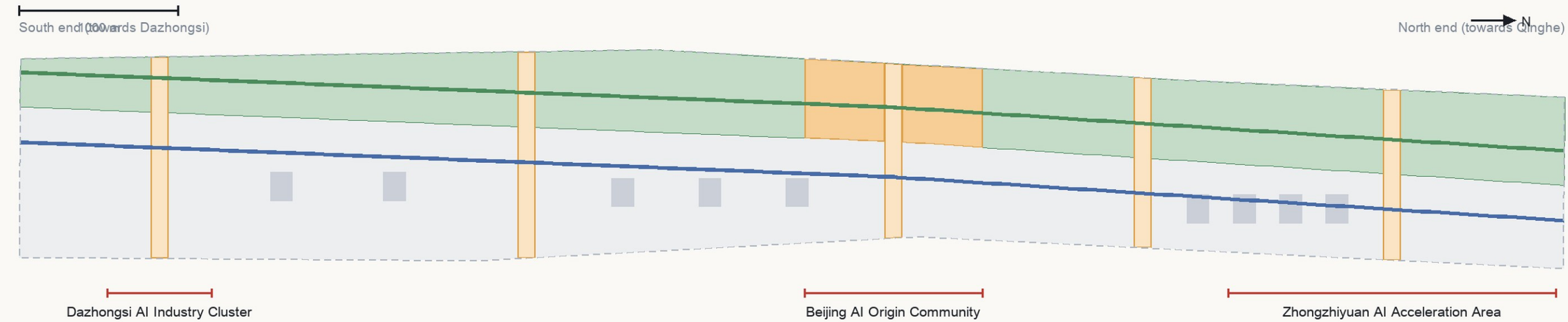
Provisional extent, not a precise-area basis

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Dazhongsi is the descending leg, the AI Origin Community the apex, Zhongzhiyuan the climbing leg; the three are threaded by the Zhongguancun service wing and the Xiaoyuehe scenario wing rather than joined end to end, so the belt reads as one continuously walkable innovation chain.

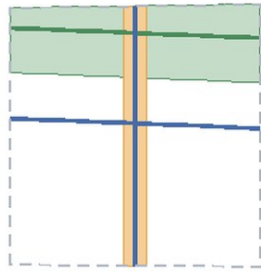
Mobility, Blue-Green and Public Space

Continuous north-south, stitched east-west · lines total approx. 25.3 km · green 34.0% · public space 10.0%



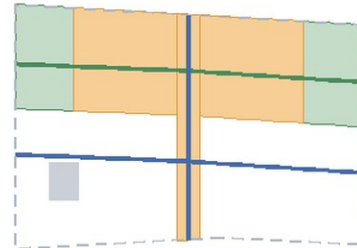
dazhongsi ai industry cluster

South: stitch streets meet the commercial hinterland



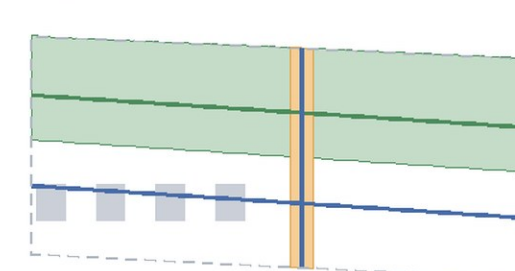
beijing ai origin community

Middle: the spine widens into the apex plaza



zhongzhiyuan ai acceleration area

North: spine directly abuts research clusters



- West slow-mobility spine (never cut by motor traffic)
- East scenario service line
- East-west stitch streets (5)
- Green and open space
- Public space: apex plaza and stitch corridors
- Illustrative footprints (not a development-scale conclusion)
- Provisional rough boundary (dashed)

Four spatial strategies

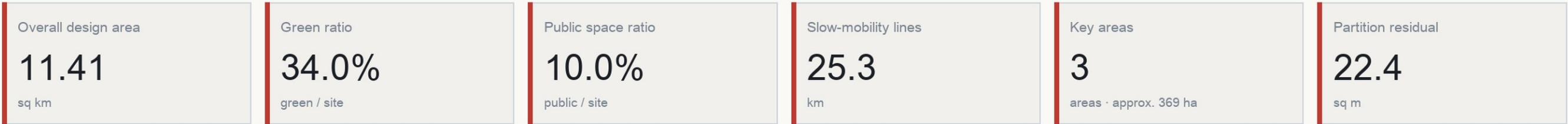
- Continuous north-south**
The 9.7 km west spine is continuous, shared by walking, cycling and low-speed autonomous delivery.
- Stitched east-west**
Five transverse corridors align to the three key areas and two intermediate joints, stitching the long-separated sides.
- Reversing apex**
At the AI Origin Community the spine widens into the plaza and the east line yields, forming the belt's civic high point.

Data: geometry from geometry/*_geojson, areas recomputed in EPSG:4548; boundary from the maintainer-registered provisional rough boundary brief/site-package/geometry/provisional_boundaries.geojson. The map is rotated 90 degrees clockwise; see the north arrow. Dashed lines are a provisional constraint. They are not an official redline, an approval basis, or a precise-area basis; recompute everything once official polygons are released. An open co-creation concept suggestion. It does not replace statutory planning; floor area ratio, height, demolition decisions and road redlines are outside its conclusions.

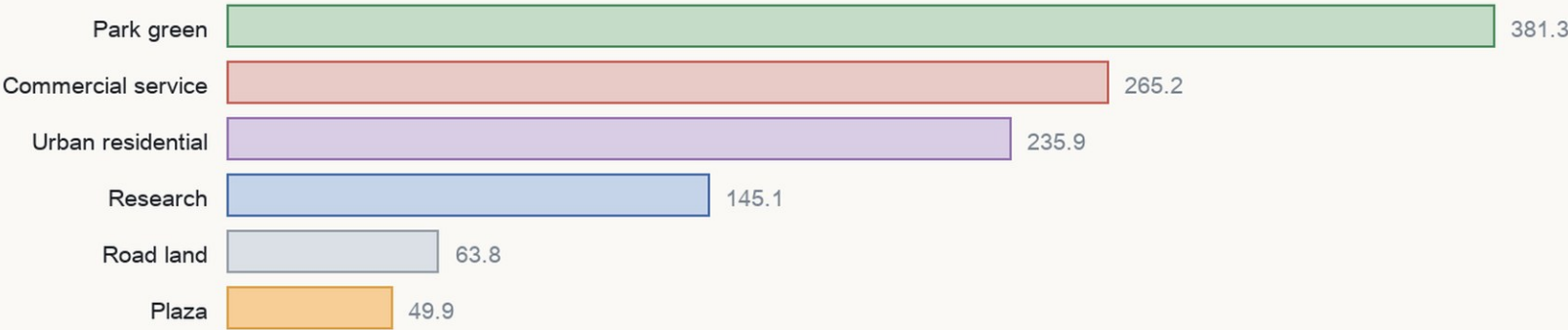
The west spine is the belt's only 9.7 km continuous slow-mobility space never cut by motor traffic; all east-west vehicular movement is organised onto the five stitch streets, whose crossings with the spine are suggested to be handled with pedestrian priority.

Metrics and Evidence Chain

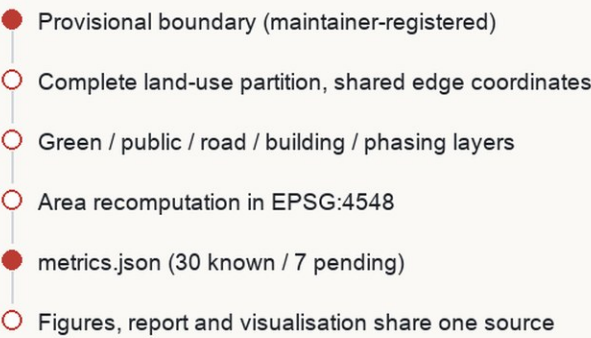
Every known metric is recomputed from geometry in EPSG:4548 and regenerates from scratch · missing statutory controls are declared pending



Land-use composition (hectares)



Evidence chain



Statutory controls pending official data (an agent does not conclude these)

- ☐ Total floor area

pending official data
- ☐ Floor area ratio

pending official data
- ☐ Building height

pending official data
- ☐ Green ratio control

pending official data
- ☐ Building density control

pending official data
- ☐ Setback requirement

pending official data

Data: geometry from geometry/*.geojson, areas recomputed in EPSG:4548; boundary from the maintainer-registered provisional rough boundary brief/site-package/geometry/provisional_boundaries.geojson. The map is rotated 90 degrees clockwise; see the north arrow. Dashed lines are a provisional constraint. They are not an official redline, an approval basis, or a precise-area basis; recompute everything once official polygons are released. An open co-creation concept suggestion. It does not replace statutory planning; floor area ratio, height, demolition decisions and road redlines are outside its conclusions.

All known metrics are recomputed from geometry in EPSG:4548 and the whole table regenerates from build/gen_metrics.py; six statutory controls are explicitly registered as unknown rather than replaced by estimates.

Metric set and area recalculation

30 recomputed known metrics / 6 explicitly registered unknown statutory controls / all derived from geometry in EPSG:4548

Metric	Recomputed
Design area	1,141.3 ha
Land-use partition total	1,141.3 ha
Partition residual	22.399 m²
Park green 1401	381.3 ha
Plaza 1403	49.9 ha
Research 0802	145.1 ha
Urban residential 0701	235.9 ha
Commercial service 05	265.2 ha
Road land 1207	63.8 ha
Green space area	387.7 ha
Green ratio	34.0%
Public space area	113.7 ha
Public space share	10.0%
Road land area	63.8 ha

Metric	Recomputed
Road land share	5.6%
Road centreline length	25,252 m
Building footprint	22.7 ha
Footprint density	2.0%
Illustrative clusters	9
Key areas total	369.3 ha
Zhongzhiyuan	192.9 ha
AI Origin Community	104.3 ha
Dazhongsi	72.0 ha
Phasing total	561.5 ha
Phase 1 (middle)	172.9 ha
Phase 2 (north)	249.8 ha
Phase 3 (south)	138.9 ha

Statutory controls explicitly registered as unknown (never replaced by estimates)

Total floor area	Floor area ratio	Building height	Official green-ratio control	Official density control	Official setback
status: unknown	status: unknown	status: unknown	status: unknown	status: unknown	status: unknown

Recomputation contract: delete metrics.json, rerun build/gen_metrics.py, and the numbers come back identical. Geometry is exchanged in EPSG:4326 with [lon, lat] ordering while areas are computed in the EPSG:4548 projected system. The text, the drawings and the offline HTML exhibition read the same metrics.json, so the three sets of numbers cannot contradict one another.

Boundary statement, matters left unconcluded, and copyright

What makes agent output usable is what it explicitly refuses to claim

Spatial base precision

The official site boundary and the three key-area polygons have not been released. Every spatial conclusion here rests on the maintainer-registered provisional rough boundary, labelled `official_boundary=false`, `geometry_role=provisional_constraint`, `boundary_precision=provisional_rough`. It supports generation, self-check, visualisation and design discussion only, never an official redline, an approval basis or a precise-area basis. Once official polygons are released the package must be recomputed as a whole rather than patched file by file.

Matters left unconcluded

Regulatory adjustment, floor area ratio, building height and development intensity and other statutory planning judgements; parcel-level retain-renovate-demolish schemes; road alignment, rail alignment, bridge and tunnel works and utility networks; underground engineering feasibility, energy load and municipal capacity; land ownership, investment estimation, development sequencing and approval judgements. All are left to bodies with statutory authority and to professional teams.

Generation and copyright

This booklet was generated by an agent from the public announcement, the maintainer-registered public site package, and public scripts in the repository. All figures are drawn programmatically by `build/gen_figures.py` from `geometry/*.geojson` and `metrics.json`, with no third-party image, trademark, embedded typeface, portrait or paper figure used; the text is original. Buildings and scenarios in the figures are illustrative generated content and must not be used as field-survey evidence. Submitted under `COMMUNITY-DISPLAY-ONLY`.

Ethics and the boundary of human judgement

No scenario performs facial recognition or individual tracking, none collects unnecessary personal data, and each provides an opt-out alternative; every output affecting the public passes human review before execution. A test scenario is not approved operation. Agent output can be filtered and ranked, but final judgement rests with people and professional teams.

Task coverage, self-check, and layer inventory

Announcement 1.3-1.5 and agent.1-agent.6 mapped to chapters, layers and metrics

Task item	Where it is carried
1.3 Coordinated research area	Coordinated Research Area chapter
1.4 Overall design area	Overall Design Area chapter + land_use / roads / green_space layers
1.5 Key areas	Detailed Design of Key Areas + key_areas.geojson
agent.1 naming and identity	Naming system and visual identity direction
agent.2 AI scenarios	Ten scenario cards + three industry test scenarios
agent.3 ecosystem and personas	Eight-factor ecosystem map + five personas
agent.4 global cases	Six publicly documented park mechanisms
agent.5 cultural narrative	Three threads: heritage, Zhongguancun, new AI culture
agent.6 events and operations	One principal, four seasonal, continuous events + operations

Geometry layers (EPSG:4326 exchange, EPSG:4548 recalculation)

site_boundary.geojson	key_areas.geojson	land_use.geojson	green_space.geojson	public_space.geojson	roads.geojson	buildings.geojson	phasing.geojson	constraints.geojson
1 features	3 features	20 features	6 features	6 features	7 features	9 features	3 features	5 features

Four classes of self-check: scripts/validate_submission.py for deterministic validation, scripts/spatial_review.py for spatial consistency, scripts/visual_review.py for exhibition packaging, and scripts/professional_review.py for the professional evidence chain. All four must pass before review. This booklet, the offline HTML exhibition, the GeoJSON layers and the metric table are evidence for one another, and any part can be replaced and rechecked.